Product Requirements Document (PRD)

Fish Pond Management System (FPMS)

Version: 3.0 (Revised MVP)
Status: MVP Scope
Platform: Web Application

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1. Product Overview

Purpose

The Fish Pond Management System (FPMS) is a web-based farm management platform that enables fish farmers to manage their entire farm from one centralized dashboard.

The platform focuses on:

- Financial management
- Pond management
- Feed inventory
- Feeding documentation
- Production cost tracking
- Notifications and reminders

The system is designed to reduce manual record keeping, improve inventory accuracy, minimize feed wastage, and provide complete operational visibility.

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2. Product Goals

Primary Goals

- Digitize farm operations
- Track financial performance
- Manage ponds efficiently
- Track feed inventory accurately
- Record daily feeding activities
- Reduce inventory errors
- Improve profitability
- Generate operational insights

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3. User Roles

Admin

Permissions

- Full system access
- Manage ponds
- Manage inventory
- Record financial data
- View reports
- Configure notifications

Employee

Permissions

- Log feeding
- Open feed bags
- Record mortality
- View assigned ponds

Restrictions

- Cannot delete records
- Cannot modify financial records

Viewer

Permissions

- Read-only access

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4. Product Modules

The MVP consists of six modules.

1. Financial Dashboard
2. Pond Management
3. Feed Inventory
4. Feed Documentation

5. Pricing
6. Notifications

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5. Financial Dashboard

Objective

Provide a complete financial overview of the farm.

Dashboard Metrics

Display

- Total Revenue
- Total Expenses
- Net Profit
- Feed Costs
- Fish Stock Costs
- Maintenance Costs
- General Overhead
- Inventory Value
- Total Ponds
- Total Fish Population

Financial Actions

Add Expense

Fields

- Category
- Amount
- Date
- Pond (Optional)
- Description

Expense Categories

- Feed
- Fish Stock
- Maintenance
- Medication

- Utilities
- Labor
- Transportation
- General Overhead
- Miscellaneous

Add Revenue

Fields

- Revenue Source
- Amount
- Date
- Notes

Revenue Sources

- Fish Sales
- Pond Rental
- Other Income

Financial Analytics

Display

- Cost vs Revenue Graph
- Monthly Profit Trend
- Expense Breakdown
- Revenue Breakdown

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6. Pond Management

Objective

Manage every pond and all activities associated with it.

Pond Overview

Display

- Total Ponds
- Active Ponds
- Empty Ponds
- Total Fish Population
- Aggregate Pond Costs

Pond List

Each card displays

- Pond Name
- Pond ID
- Pond Type
- Current Fish Count
- Pond Status
- Total Cost

Add New Pond

Fields

- Pond Name
- Pond ID
- Pond Type
- Pond Size
- Water Capacity
- Maximum Stock Capacity
- Initial Fish Stock
- Average Fish Weight
- Stocking Date
- Notes

Pond Details

The Pond Details page becomes the operational dashboard for an individual pond.

Fish Information

Display

- Initial Fish Stock
- Current Fish Count
- Average Fish Weight
- Stocking Date
- Current Mortality
- Mortality Rate

Mortality

Button

Log Mortality

Fields

- Date
- Number of Dead Fish
- Cause
- Notes

Mortality Logic

Current Fish Count

=

Current Fish Count

−

Dead Fish

Mortality Rate

=

(Total Dead Fish ÷ Initial Fish Stock)

×

100

The system updates automatically

- Current Fish Count
- Mortality Percentage
- Pond Statistics

Cost Tracking

Button

Add Pond Cost

Fields

- Category
- Amount
- Date
- Description

Feed Summary

Automatically grouped by Brand and Feed Size.

Example

Brand	Size	Total Feed Given
Coppens	2mm	24kg
Coppens	4mm	75kg
Aqua	6mm	112kg

These totals are generated automatically from Feeding Documentation.

No manual editing.

Feeding History

Display

- Date
- Feed Brand
- Feed Size
- Morning Feed
- Evening Feed
- Total Feed
- Employee

Example

12 June

Coppens

2mm

Morning

2kg

Evening

1kg

Total

3kg

Pond Performance

Display

- Fish Growth
- Mortality Trends
- Feed Consumption
- Cost Trends

Daily Feeding Calculator

Purpose

Recommend daily feed requirements.

Inputs

- Fish Count
- Average Weight
- Feeding Rate

Outputs

- Recommended Feed Size
- Morning Feed
- Evening Feed
- Daily Feed Required

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7. Feed Inventory

Objective

Manage unopened feed stock.

Inventory and feeding are intentionally separated.

Inventory Summary

Display

- Total Bags
- Total Kilograms
- Opening Inventory
- Feed Purchased
- Feed Issued (Opened Bags)
- Closing Inventory

Feed Purchase

Fields

- Purchase Date
- Feed Brand
- Feed Size
- Bags Purchased
- Weight Per Bag
- Cost Per Bag
- Supplier

Open Feed Bags

Purpose

Issue feed from inventory into operational use.

This is the ONLY place inventory is deducted.

Log Open Bags

Fields

- Date
- Feed Brand
- Feed Size
- Number of Bags Opened
- Weight Per Bag
- Opened By

System Logic

When submitted

1.

Verify available bags

↓

2.

Deduct opened bags from inventory

↓

3.

Create Open Feed Batch

↓

4.

Calculate total available kilograms

Example

Inventory

Coppens

4mm

20 Bags

User Opens

3 Bags

Inventory becomes

17 Bags

Open Feed Batch

75kg

Open Feed Batches

Display

- Feed Brand

- Feed Size
- Bags Opened
- Initial Weight
- Remaining Weight
- Date Opened
- Status

Status

- Open
- Completed

Inventory History

Display

- Purchases
- Bags Opened
- Inventory Adjustments

Feed Forecast Calculator

Purpose

Estimate total feed needed until harvest.

Inputs

- Fish Count
- Current Weight
- Target Weight
- FCR
- Feed Bag Weight
- Cost

Outputs

- Feed Required (kg)
- Bags Required
- Estimated Cost
- Monthly Feed Plan

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8. Feed Documentation

Objective

Document every feeding event.

Feed Documentation DOES NOT deduct inventory.

It deducts only from Open Feed Batches.

Log Feeding

Fields

- Date
- Pond
- Open Feed Batch
- Feed Brand
- Feed Size
- Morning Feed
- Evening Feed
- Notes

Feeding Logic

System

Calculate

Morning +

Evening

=

Total Feed

↓

Deduct from Open Feed Batch Remaining Weight

↓

Update Remaining Weight

↓

Create Feeding Record

↓

Update Pond Feed History

↓

Update Pond Feed Summary

Example

Open Feed Batch

75kg

Morning

8kg

Evening

10kg

Remaining

57kg

Inventory stays unchanged.

Pond Integration

Every feeding record automatically updates the selected pond.

The system maintains cumulative feed totals grouped by:

- Feed Brand
- Feed Size

Example

Today

Pond 1

Coppens

2mm

2kg

Tomorrow

Pond 1

Coppens

2mm

4kg

System automatically updates

Coppens

2mm

6kg Total

Feeding History

Display

- Date
- Pond
- Feed Brand
- Feed Size
- Morning Feed
- Evening Feed
- Total Feed
- Recorded By

Filters

- Pond
- Brand
- Feed Size
- Date Range

Outputs

- Daily Feed Usage
- Feed History
- Feed Usage by Pond
- Feed Usage by Brand
- Feed Usage by Feed Size
- Remaining Open Feed Stock

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9. Pricing

Objective

Calculate production costs and profitability.

Display

- Cost Per Fish
- Cost Per Pond
- Feed Cost Allocation
- Operational Cost Allocation
- Expected Revenue
- Expected Profit
- Suggested Selling Price
- ROI

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10. Notifications

Feeding

- Morning Reminder
- Evening Reminder

Inventory

- Low Stock
- Reorder Alert
- Out of Stock

Pond

- Mortality Alert
- Sampling Reminder
- Harvest Reminder

Financial

- High Expense Alert
- Monthly Summary
- Revenue Performance Alert

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11. Data Relationships

The platform follows one source of truth.

Feed Purchase

↓

Feed Inventory

(Unopened Bags)

↓

Open Feed Bags

↓

Open Feed Batch

↓

Feed Documentation

↓

Pond Feed History

↓

Pond Feed Summary

This ensures:

- Inventory is deducted only once.
- Feeding history remains accurate.
- Every pond has a complete feeding record.
- Remaining feed can always be tracked.

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12. Business Rules

Inventory Rules

- Inventory stores unopened bags only.
- Feeding never deducts inventory.
- Only "Open Feed Bags" deduct inventory.

Open Feed Rules

- Every opened bag creates an Open Feed Batch.
- Multiple Open Feed Batches can exist simultaneously.
- Feed is consumed only from an Open Feed Batch.
- When Remaining Weight reaches 0kg, the batch status changes to Completed.

Feeding Rules

- Every feeding record must reference an Open Feed Batch.
- Every feeding record must reference a Pond.
- Feed records automatically update Pond Feed Summary.
- Feed records automatically update Pond Feed History.

Pond Rules

- Every pond maintains its own feed history.

- Feed totals are grouped by Feed Brand and Feed Size.
- Mortality automatically updates Current Fish Count.
- Current Fish Count can never exceed Initial Fish Stock.

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13. Future Roadmap

Phase 2

- Water Quality Monitoring
- AI Feed Recommendations
- Growth Forecasting
- WhatsApp Notifications

Phase 3

- Multi-Farm Management
- Fish Marketplace
- Veterinary Module
- Market Price Intelligence

MVP Scope

The first release includes:

- Financial Dashboard
- Pond Management
- Feed Inventory
- Open Feed Bag Management
- Feed Documentation
- Daily Feeding Calculator
- Feed Forecast Calculator
- Pricing
- Notifications

This architecture provides a scalable foundation for fish farm operations while ensuring inventory accuracy, traceable feeding records, and clear separation between stock management and daily farm activities.